

Technical Report - Parkrun Data Analysis

Group H Lab 02

2025-10-10

Contents

1	Contributions	2
2	Introduction	3
3	Data Pipeline Overview	3
3.1	Understanding the data	3
3.2	Data Cleaning	3
3.3	Adding New Variables	4
4	Analysis Questions	4
5	Limitations	4
6	Recommendations	5
7	Conclusion	5

1 Contributions

This project was completed by our team with the following roles:

Raman Preet Singh:

- Set up Git repository and created the Trello dashboard.
- Managed and regularly updated the progress in Trello dashboard.
- Started with data cleaning by importing data files, cleaning messy datasets.
- Continued with the analysis from 4 to 6 questions.
- Finalised the charts and maps used to present the findings and completed the stakeholder's report.
- Contributed to README.

Rupika Sivasubramanian:

- Continued with the data cleaning.
- Created final dataset for analysis.
- Worked with the analysis questions 1 to 3.
- Worked on technical report.
- Contributed to README.

Huizhi Zhang:

- Contributed to the technical report.

2 Introduction

This report presents a technical analysis of parkrun data from seven Ōtautahi locations to help Puma spot the best marketing opportunities, understand how runners behave, and better target potential customers.

3 Data Pipeline Overview

3.1 Understanding the data

The raw data came as multiple CSV files. Each file contained results from one parkrun event. The files were named following this pattern: {location}*parkrun*{event_id}.csv. Analysed data from seven parkrun locations:

- Hagley
- Broadpark
- Foster
- Halswell Quarry
- Pegasus
- Riverlution
- Scarborough

From these files, following variables were extracted for the final analysis:

- id: Unique identifier for each runner
- week: Week number
- parkrun: Location name
- sex: Gender
- age: Age group
- club: Running club name
- time: Finish time

3.2 Data Cleaning

Importing files: Used the `map_dfr()` function to import all CSV files and combine them into one dataset. Added a filename column and extracted the parkrun location name and parkrun week number.

Splitting the parkrunner column: The parkrunner column contained all required information separated by “\n” (line breaks), which was split it into 9 separate columns and extracted: parkrunner name and total parkruns completed from `coll`; age category and age grading percentage from Age Group using regular expressions; gender value from Gender.

Creating age categories: After extracting the age groups, grouped them into 3 simple categories - Junior (Age 19 and under), Senior (Age 20 to 44), and Veteran (Age 45 and over).

Handling missing values: Parkrunners without names were removed completely. Missing `age_group`, `club`, or `gender` were kept but marked as “Not Available”. Missing age grading values were set to 0%.

Cleaning time values: Split the Time column into Time1 (race finish time) and Time2 (remaining text). Extracted `Personal_Best_Time` from Time2 if present, otherwise used Time1 as the default. Converted all times to proper time format.

Basic text cleaning: Removed extra spaces from the beginning and end of all text. Made parkrunner names look proper by capitalizing each word.

3.3 Adding New Variables

Milestone Club: Based on the total number of parkruns completed, participants were assigned to milestone clubs.

- Under 25 parkruns (under-18): “Member of the 10 Club” (juniors Only)
- 25-49 parkruns: “Member of the 25 Club”
- 50-99 parkruns: “Member of the 50 Club”
- 100-249 parkruns: “Member of the 100 Club”
- 250-499 parkruns: “Member of the 250 Club”
- 500+ parkruns: “Member of the 500 Club”

Unique parkrunner ID: Assigned unique parkrunner_id to each runner starting from 10000 by getting the unique runners from parkrun name and age group combination.

event_id Weeks were numbered from 1 to 11 for easy comparison of weekly trends across all parkruns.

4 Analysis Questions

Variables and datasets created for analysis questions are discussed below

Questions 1 & 3: For Q1, calculated unique_runners (count of unique runners per location), unique_runners_per_week (weekly participation by location), and weekly_average. For Q3 and Q6, created parkrun_count (number of different parkruns each person attended), an overlap_matrix (showing shared participants between parkrun pairs), overlap_runners (count attending both parkruns), union (total unique runners across both parkruns), and mixing_pct (overlap percentage calculated as $\text{overlap} / \text{union} \times 100$).

Question 2: Created first_week (each runner’s first participation each parkrun), runner_tag (whether “New” vs “Returning”), new_runners (count of first-time participants per week), returning_runners (count of repeat participants per week).

Questions 4 & 5: For Q4, created cutoff_time (10th percentile finish time per parkrun as a location-specific threshold), elite (binary indicator where 1=elite, 0=non-elite based on finishing in top 10%), elite_runners (count of elite runners per location), and weekly_elites (weekly count of elite participants). For Q5, created cnt (number of weeks attended per runner per location), attended (categorical variable: “Only 1 week”, “2-5 weeks”, “5+ weeks”), consistent (binary indicator where 1=attended 6+ weeks, 0=otherwise), consistent_runners (count of consistent participants per location).

Question 6: Analysed the relationship between physical distance and runner mixing patterns and created parkrun_dist (a dataframe containing latitude and longitude coordinates for each location), distance_km (Haversine distance in kilometers between each parkrun pair), calculated the Pearson correlation coefficient between distance and mixing percentage, and built a log regression model $\text{lm}(\text{mixing_pct} \sim \log(\text{distance_km}))$ to assess the relationship between distance and mixing behavior.

5 Limitations

Week Some parkruns, like Hagley, have been around for a long time and naturally have more participants. Others, like Riverlution, are newer parkruns. So, this is a limitation when comparing participation just by weekly data without considering the start date.

Data Missingness Parkrunner Names were missing which had to be ignored which is the basic of the analysis and high percentage amount of records lack age group, gender, or club information, limiting demographic analysis.

Club There is missingness and inconsistency in Club also there was no clarity on which club data has to be considered for the analysis, also many club member details were missing.

Elite Runners There is currently no clear definition for elite or consistent runners, so we have used finish time as the main criteria. However, having a proper definition would help improve the analysis - whether it's based on age, finish time, or a combination of both.

6 Recommendations

Improved data collection Make sure runners (returning and new) fill in all important details like age, gender, and club when they sign up. This will reduce the many “Not Available” entries and having complete information will help us understand runners’ details better.

Elite runners Classification Instead of just using finish time to identify elite runners, we should combine multiple factors like finish time, age-grading percentage, and how consistently they participate. This would give us a better “elite score” that considers both performance and commitment, making it easier for Puma to target the right runners.

Holidays Information: Including details about holidays could be helpful to check if participation is higher on those days and it can be an attribute to check the trends in increased no of runners.

Runner’s Unique ID Instead of creating IDs by combining names and age groups (which can change over time or be the same for different people), we should use the existing parkrun ID for each runner. To protect privacy, this ID can be converted into a secure, anonymous code using hashing or masking anything similar. This way, each runner keeps the same anonymous ID across all events and locations, making the data more accurate and consistent while still keeping personal information private.

7 Conclusion

This analysis of parkrun data provides details about runner behavior, participation trends, and demographic patterns. The report helps Puma connect with the runner community and make strategic decisions that maximize business profits.